This is particularly useful as

$$\begin{aligned}[\mathcal{L}_{\mathrm{A}}^{\dagger}, \mathcal{L}_{\mathrm{B}}^{\dagger}]\rho_\beta &= \left(\mathcal{L}_{\mathrm{A}}^{\dagger}\mathcal{L}_{\mathrm{B}}^{\dagger} - \mathcal{L}_{\mathrm{B}}^{\dagger}\mathcal{L}_{\mathrm{A}}^{\dagger}\right)\rho_\beta = -\left(\mathcal{L}_{\mathrm{A}}^{\dagger} + \mathcal{L}_{\mathrm{B}}^{\dagger}\right)\left(g(\boldsymbol{q},\boldsymbol{p})\rho_\beta\right) \\ &= \left(\mathcal{L}_{\mathrm{O}}^{\dagger} - \mathcal{L}_{\mathrm{LD}}^{\dagger}\right)\left(g(\boldsymbol{q},\boldsymbol{p})\rho_\beta\right).\end{aligned}$$

Now notice that for a function $\psi(\boldsymbol{q})$ purely of position, we have

$$\mathcal{L}_{\mathrm{LD}}^{\dagger}\left(\psi(\boldsymbol{q})\rho_\beta\right) = -\left(\boldsymbol{p}^T\boldsymbol{M}^{-1}\nabla\psi(\boldsymbol{q})\right)\rho_\beta(\boldsymbol{q},\boldsymbol{p}),$$

and hence we can compute

$$\begin{aligned}\mathcal{L}_{\mathrm{O}}^{\dagger}\left(g(\boldsymbol{q},\boldsymbol{p})\rho_\beta\right) &= \gamma\rho_\beta\left(-p\cdot\nabla_p g(\boldsymbol{q},\boldsymbol{p}) + \beta^{-1}\nabla_p\cdot\boldsymbol{M}\nabla_p g(\boldsymbol{q},\boldsymbol{p})\right), \\ &= -\gamma g(\boldsymbol{q},\boldsymbol{p})\rho_\beta, \\ &= \gamma\beta\mathcal{L}_{\mathrm{LD}}^{\dagger}\left(U(\boldsymbol{q})\rho_\beta\right).\end{aligned}$$

This means that

$$\mathcal{L}_{1,[\![\mathrm{ABO}]\!]}^{\dagger}\rho_\beta = -\frac{1}{2}[\mathcal{L}_{\mathrm{A}}^{\dagger}, \mathcal{L}_{\mathrm{B}}^{\dagger}]\rho_\beta = -\frac{1}{2}\mathcal{L}_{\mathrm{LD}}^{\dagger}\left(\left[\gamma\beta U(\boldsymbol{q}) - g(\boldsymbol{q},\boldsymbol{p})\right]\rho_\beta\right).$$

Hence, we can combine this equation with (7.21) to yield

$$\mathcal{L}_{\mathrm{LD}}^{\dagger}f_{1,[\![\mathrm{ABO}]\!]}\rho_\beta = \frac{1}{2}\mathcal{L}_{\mathrm{LD}}^{\dagger}\left(\left[\gamma\beta U(\boldsymbol{q}) - g(\boldsymbol{q},\boldsymbol{p})\right]\rho_\beta\right),$$

with the general solution

$$f_{1,[\![\mathrm{ABO}]\!]} = \frac{\beta}{2}\left(\gamma U(\boldsymbol{q}) - \boldsymbol{p}^T\boldsymbol{M}^{-1}\nabla U(\boldsymbol{q}) - c\right),$$

where $c = \gamma\langle U(\boldsymbol{q})\rangle$ is a constant, chosen to ensure that the average of the correction function is zero. This solution gives the leading order correction for the three related schemes in (7.23).

For the final two schemes, we can compute the correction function through the use of (7.24). Our previous work has computed the second term on the right hand side already, so we are left with the initial term. However, its expression reduces quite easily, as

$$\begin{aligned}[\mathcal{L}_{\mathrm{A}}^{\dagger} - \mathcal{L}_{\mathrm{B}}^{\dagger}, \mathcal{L}_{\mathrm{O}}^{\dagger}]\rho_\beta &= -\mathcal{L}_{\mathrm{O}}^{\dagger}\left(\mathcal{L}_{\mathrm{A}}^{\dagger} - \mathcal{L}_{\mathrm{B}}^{\dagger}\right)\rho_\beta, \\ &= -2\mathcal{L}_{\mathrm{O}}^{\dagger}\left(g(\boldsymbol{q},\boldsymbol{p})\rho_\beta\right), \\ &= -2\gamma\beta\mathcal{L}_{\mathrm{LD}}^{\dagger}\left(U(\boldsymbol{q})\rho_\beta\right).\end{aligned}$$

And hence we have

$$\mathcal{L}_{\mathrm{LD}}^{\dagger}\left(f_{1,[\![\mathrm{AOB}]\!]}\rho_\beta\right) = \mathcal{L}_{\mathrm{LD}}^{\dagger}\left(f_{1,[\![\mathrm{ABO}]\!]}\rho_\beta\right) - \gamma\beta\mathcal{L}_{\mathrm{LD}}^{\dagger}\left(U(q)\rho_\beta\right),$$

giving the general solution

$$f_{1,[\![\mathrm{AOB}]\!]} = -\frac{\beta}{2}\left(\gamma U(q) + \boldsymbol{p}^T\boldsymbol{M}^{-1}\nabla U(\boldsymbol{q}) - c\right),$$

for the same choice of c.

Hence using the symmetry in the commutators we are able to solve for the leading order corrections for each of the schemes of the form $[\![\mathrm{XYZ}]\!]$. In summary, we find that

$$f_{1,[\![\mathrm{ABO}]\!]}(\boldsymbol{q},\boldsymbol{p}) = f_{1,[\![\mathrm{OAB}]\!]}(\boldsymbol{q},\boldsymbol{p}) = \frac{\beta}{2}\left(\gamma U(\boldsymbol{q}) - \boldsymbol{p}^T\boldsymbol{M}^{-1}\nabla U(\boldsymbol{q}) - c\right), \tag{7.25a}$$

$$f_{1,[\![\mathrm{OBA}]\!]}(\boldsymbol{q},\boldsymbol{p}) = f_{1,[\![\mathrm{BAO}]\!]}(\boldsymbol{q},\boldsymbol{p}) = -\frac{\beta}{2}\left(\gamma U(\boldsymbol{q}) - \boldsymbol{p}^T\boldsymbol{M}^{-1}\nabla U(\boldsymbol{q}) - c\right), \tag{7.25b}$$

$$f_{1,[\![\mathrm{AOB}]\!]}(\boldsymbol{q},\boldsymbol{p}) = -\frac{\beta}{2}\left(\gamma U(\boldsymbol{q}) + \boldsymbol{p}^T\boldsymbol{M}^{-1}\nabla U(\boldsymbol{q}) - c\right), \tag{7.25c}$$

$$f_{1,[\![\mathrm{BOA}]\!]}(\boldsymbol{q},\boldsymbol{p}) = \frac{\beta}{2}\left(\gamma U(\boldsymbol{q}) + \boldsymbol{p}^T\boldsymbol{M}^{-1}\nabla U(\boldsymbol{q}) - c\right), \tag{7.25d}$$

where the constant c is chosen to ensure each correction function has average 0, giving $c = \gamma\langle U(\boldsymbol{q})\rangle$. If required, we may continue this procedure in order to find correction functions for higher orders, though the added complexity means that we cannot always solve to find an explicit solution.

The results of our analysis suggest that to leading order the error scales linearly with the reciprocal temperature β, but more importantly a non-constant term appears which is of order γ. As our correction is of the form $\langle\vartheta f_1\rangle$ we can see that if $\vartheta = \vartheta_1(\boldsymbol{p})$ then this $\mathcal{O}(\gamma)$ term will not trouble us, as

$$\begin{aligned}\langle\vartheta_1(\boldsymbol{p})f_1(\boldsymbol{q},\boldsymbol{p})\rangle &= \pm\frac{\beta}{2}\left\langle(\gamma U(\boldsymbol{q}) - c)\vartheta_1(\boldsymbol{p}) \pm \boldsymbol{p}^T\boldsymbol{M}^{-1}\nabla U(\boldsymbol{q})\vartheta_1(\boldsymbol{p})\right\rangle \\ &= \pm\frac{\beta}{2}\left\langle\boldsymbol{p}^T\boldsymbol{M}^{-1}\nabla U(\boldsymbol{q})\vartheta_1(\boldsymbol{p})\right\rangle,\end{aligned}$$

where the sign depends on the particular scheme used (which governs the form of f_1). Thus in the case of a function of pure momentum the order γ term does not